CUSTOMER SHOPPING BEHAVIOUR ANALYSIS

A Data Analytics Project Report

PostgreSQL • Python (Pandas) • Power BI

Dataset: 3,900 customer shopping records | Total recorded purchase value: $233,081

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Executive Summary

This project analyses customer shopping behaviour using 3,900 records. The workflow combines Python-based data preparation, PostgreSQL queries for business questions, and a Power BI dashboard for management-friendly visual exploration. The dataset contains customer demographics, purchased items and categories, purchase amounts, review ratings, subscription status, shipping type, discount usage, previous purchases, payment method, and purchase frequency.

At a glance	Result
Customer records	3,900
Total recorded purchase value	$233,081
Average purchase amount	$59.76
Average review rating	3.75 / 5
Subscribed customers	1,053 (27.0%)
Largest revenue category	Clothing — $104,264
Highest-revenue gender group	Male — $157,890

Key takeaways

- Clothing contributes the largest recorded revenue ($104,264), followed by Accessories ($74,200).
- Male customers account for $157,890 in recorded purchase value; female customers account for $75,191. This describes this dataset and does not imply a general market rule.
- 27.0% of records are marked subscribed. Average spend is very similar for subscribed and non-subscribed records.
- The dashboard provides quick comparisons by category, age group, subscription status, gender, and shipping type.

Interpretation note: “Revenue” in this report means the sum of the dataset’s Purchase Amount (USD) values. It should not be interpreted as audited company revenue or profit.

1. Introduction

1.1 Background

Understanding customer shopping behaviour helps retailers make better decisions about product assortment, promotions, customer retention, and service options. Data analysis can turn transaction-level records into concise evidence that a manager can use.

1.2 Project objective

- Explore the customer dataset and understand its structure and quality.
- Prepare the data for analysis using Python and Pandas.
- Use PostgreSQL to answer business questions with aggregations, filters, sorting, and ranking.
- Present the findings through a clear Power BI dashboard and plain-language conclusions.

1.3 Scope

The analysis focuses on purchase value, gender, product category and item, customer age group, subscriptions, discounts, shipping types, ratings, and previous-purchase behaviour. The dataset is observational; it supports descriptive analysis, not proof of causation.

1.4 Tools used

Tool	Role in the project
Python + Pandas	Inspecting, cleaning, transforming, and preparing data
PostgreSQL + pgAdmin	Querying the prepared customer table to answer business questions
Power BI	KPI cards, category comparisons, subscription distribution, age-group charts, and slicers
Excel/CSV dataset	Source data containing 3,900 customer records

2. Dataset Overview

The supplied file is named customer_shopping_behavior.xls, but its contents are comma-separated text (CSV), which is how it was loaded for this analysis.

Dataset property	Details
Rows	3,900 customer records
Columns	18 original fields plus a derived age_group used in analysis
Purchase amount	$ values recorded as whole numbers
Missing data	Review Rating has 37 missing values; other inspected fields have no missing values
Primary analysis fields	Gender, Item Purchased, Category, Purchase Amount, Review Rating, Subscription Status, Shipping Type, Discount Applied, Previous Purchases,

	Age

Main data fields

Field	Meaning
Customer ID	Identifier for a customer record
Age / Gender / Location	Customer profile attributes
Item Purchased / Category	Product and product grouping
Purchase Amount (USD)	Recorded value of a purchase
Review Rating	Customer review score
Subscription Status	Whether the record is marked Yes or No for subscription
Shipping Type	Shipping option selected
Discount Applied / Promo Code Used	Discount and promotion indicators
Previous Purchases / Frequency of Purchases	Purchase history and reported frequency
Payment Method / Season / Size / Color	Additional descriptive attributes

3. Data Preparation with Python

The notebook shows a Pandas workflow that reads the source data, inspects the rows and schema, checks missing values, fills missing review ratings using the median within each product category, standardises column names, renames the purchase amount field, creates age groups and a purchase-frequency-days field, removes the duplicate promo-code indicator, and loads the prepared table into PostgreSQL.

Preparation step	Purpose / note
Inspect head, info, describe	Understand sample rows, data types, and distributions
Check missing values	Identify 37 missing Review Rating values
Fill Review Rating by category median	Retain rows while using a category-specific central value; verify missing values again
Standardise column names	Use lowercase names and underscores to simplify SQL references
Rename purchase amount field	Use the concise name purchase_amount
Create age_group	Divide ages into four quantile-based groups for comparison
Create purchase_frequency_days	Map reported frequency labels to approximate day intervals
Remove promo_code_used	Notebook checks that promo_code_used matches discount_applied and drops the duplicate field
Load into PostgreSQL	Write the prepared dataframe to the customer table

Quality note: the notebook's age-group labels contain spelling variants ("Youn Adult", "Middel aged"). The report uses readable labels for presentation. Frequency mappings are approximations; "Bi-Annually" and "Annually" should be checked against the intended meaning before business use.

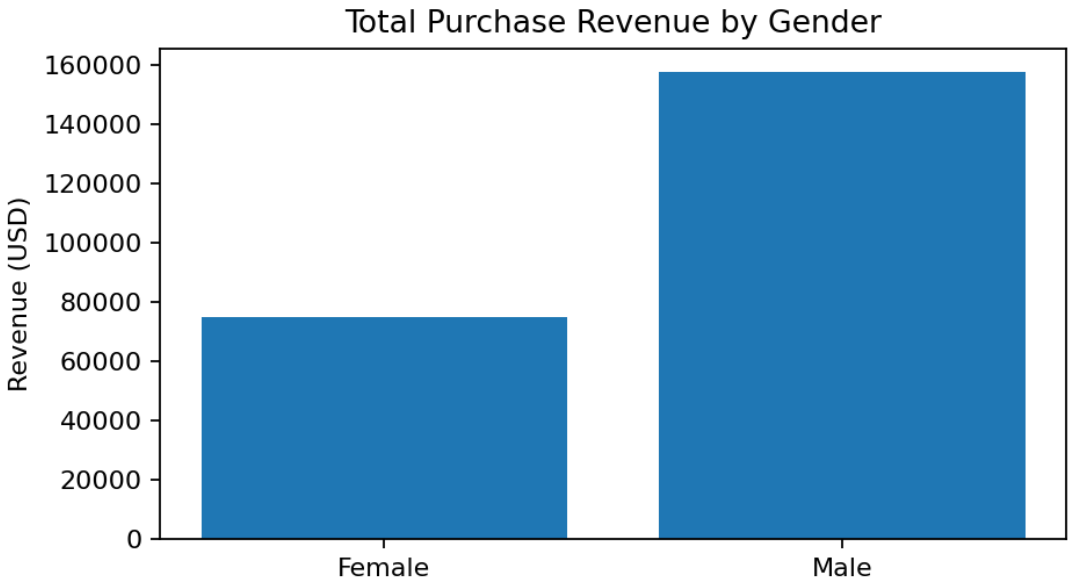
4. PostgreSQL Business Questions & Answers

The following section presents the questions, answers, and result tables in business language. SQL code is intentionally omitted so the report stays easy to read for HR, interviewers, and managers.

Q1. Which gender group contributes more purchase value?

Answer: Male customer records contribute more total purchase value in this dataset.

Gender	Customer records	Total purchase value	Average purchase amount
Female	1,248	$75,191	$60.25
Male	2,652	$157,890	$59.54



Q2. How many customers used a discount and spent above the average purchase amount?

Answer: 839 records meet both conditions: Discount Applied is “Yes” and Purchase Amount is greater than the dataset average of $59.76.

Measure	Result
Average purchase amount	$59.76
Discounted records above average	839

Q3. Which five products have the highest average review ratings?

Answer: The five highest average ratings by item and category are shown below. These are averages of the available review ratings.

Product	Category	Average review rating
Gloves	Accessories	3.863
Sandals	Footwear	3.845

Boots	Footwear	3.819
Hat	Accessories	3.801
Skirt	Clothing	3.785

Q4. How do average purchase amounts compare for Standard and Express shipping?

Answer: In these records, Express shipping has a slightly higher average purchase amount than Standard shipping. This is a descriptive comparison, not evidence that shipping type causes higher spending.

Shipping type	Records	Average purchase amount	Total purchase value
Express	646	$60.48	$39,067
Standard	654	$58.46	$38,233

Q5. Do subscribed customers spend more?

Answer: Average purchase amounts are close. Non-subscribed records have a marginally higher average purchase amount and a larger total because there are more non-subscribed records.

Subscription status	Customer records	Average purchase amount	Total purchase value
No	2,847	$59.87	$170,436
Yes	1,053	$59.49	$62,645

Q6. Which products have the highest percentage of discount usage?

Answer: Hat has the highest discount-use share among the products shown.

Product	Records with discount applied (%)
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

Q7. How many customers fall into New, Returning, and Loyal segments?

Segmentation rule used in the notebook: New = 1 previous purchase; Returning = 2–10 previous purchases; Loyal = more than 10 previous purchases.

Customer segment	Records
Loyal	3,116
Returning	701
New	83

Q8. What are the three most purchased products within each category?

Answer: The table ranks items by the number of records in which each item was purchased.

Category	Product	Purchase records	Rank
Accessories	Jewelry	171	1
Accessories	Belt	161	2
Accessories	Sunglasses	161	3
Clothing	Blouse	171	1
Clothing	Pants	171	2
Clothing	Shirt	169	3
Footwear	Sandals	160	1
Footwear	Shoes	150	2
Footwear	Sneakers	145	3
Outerwear	Jacket	163	1
Outerwear	Coat	161	2

Q9. Among repeat buyers (more than five previous purchases), how many are subscribed?

Answer: Using the question's stated definition of repeat buyer (Previous Purchases > 5), the split is:

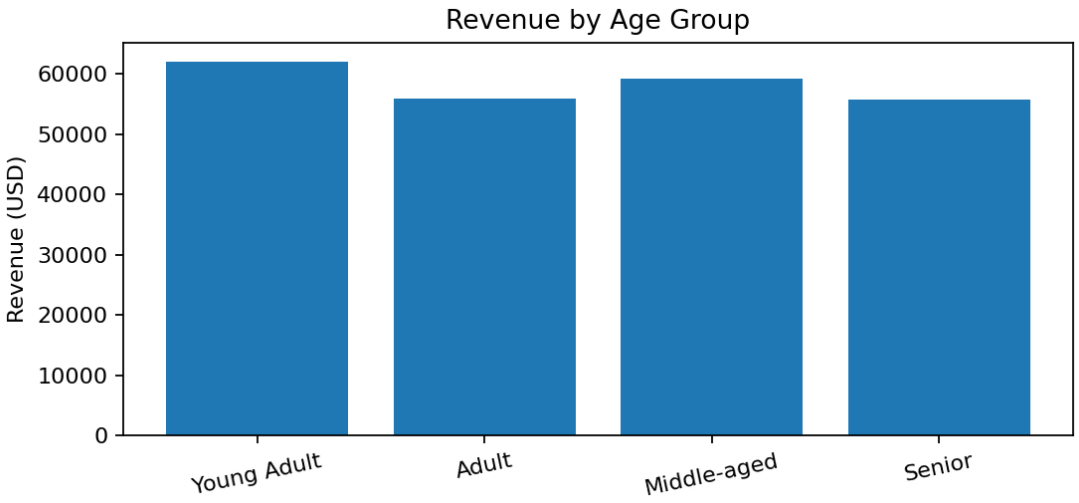
Subscription status	Repeat-buyer records
No	2,518
Yes	958

Note: The screenshot of the query appears to filter Purchase Amount >= 5, which is not the same as filtering Previous Purchases > 5. This report uses the question's definition.

Q10. What is the purchase-value contribution of each age group?

Answer: Young Adult is the highest-contributing age group by total purchase value in this dataset.

Age group	Total purchase value	Share of total
Young Adult	$62,143	26.7%
Adult	$55,978	24.0%
Middle-aged	$59,197	25.4%
Senior	$55,763	23.9%



5. Power BI Dashboard

The dashboard is designed to help a manager understand the main customer-shopping patterns at a glance. The supplied dashboard screenshot includes a customer count card, average purchase amount, average review rating, subscription-status donut chart, revenue by category, sales by category, revenue by age group, sales by age group, and slicers for subscription status, gender, category, and shipping type.

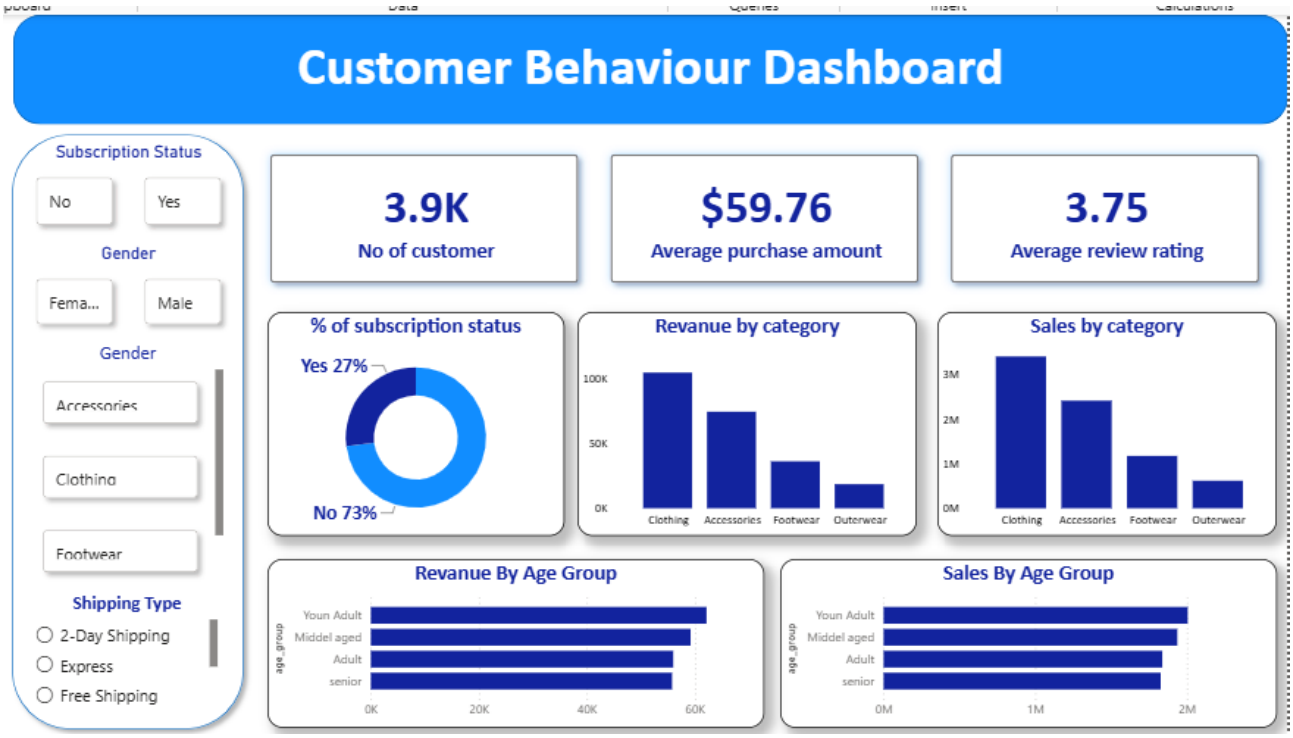


Figure 1. Customer Behaviour Dashboard supplied with the project.

How a manager can use the dashboard

- Start with the KPI cards to understand record volume, average purchase amount, and average review rating.
- Use the subscription donut to see the share of records marked subscribed versus not subscribed.
- Compare revenue and sales by category to understand which product groups contribute most.
- Compare age groups to see how recorded purchase value is distributed across customer age bands.
- Use the slicers to focus the view on a particular subscription status, gender, category, or shipping type.

Suggested improvements

- Use consistent labels: “Customer count”, “Average purchase amount (USD)”, “Average review rating”.
- Add clear currency formatting and data labels; use one consistent colour palette.
- Add a reset-filters button and tooltips that explain each chart.
- Ensure all charts respond to the slicers and add a visible “Filters applied” cue.
- Use “purchase records” instead of “sales” if each row represents one customer-level purchase and no order quantity is available.

6. Findings and Business Recommendations

Finding	Potential action
Clothing is the largest category by recorded purchase value.	Review category-level stock availability and consider featuring popular items; validate margins before investment.
Male records contribute more total purchase value, while average amounts are relatively close.	Segment campaigns using multiple variables rather than gender alone; test campaign results before scaling.
Subscription and non-subscription average purchase amounts are very similar.	Evaluate subscriptions using retention, purchase frequency, and lifetime value—not average purchase amount alone.
Discount use varies by item, with Hat highest among the top five shown.	Measure whether discounts increase conversion or merely reduce margin.
Young Adult contributes the highest total purchase value by age group.	Explore product preferences and repeat-purchase behaviour by age group before targeting.
Express shipping average purchase amount is slightly higher than Standard.	Assess delivery cost, satisfaction, and customer mix before making shipping-policy decisions.

7. Limitations

- The dataset contains 3,900 records and may not represent all customers or a real company’s full sales history.
- The analysis is descriptive; relationships in the data do not prove causation.
- The report treats each row as one purchase record, consistent with the available fields.
- The supplied dashboard displays rounded figures; minor differences may appear due to rounding or aggregation choices.
- A repeat-buyer result is calculated from Previous Purchases > 5; the screenshot query appears to use a different field in its filter.

8. Conclusion

This project demonstrates an end-to-end introductory analytics workflow: inspect and prepare customer data in Python, load it into PostgreSQL, answer practical business questions, and communicate results in Power BI. The main descriptive findings are that Clothing contributes the highest recorded purchase value, male records contribute the larger total purchase value, subscription status is associated with similar average purchase amounts, and Young Adult records contribute the most purchase value among the age groups used. The dashboard makes these patterns easier to explore through visual comparisons and filters.

Project skills demonstrated

Data inspection • Missing-value handling • Feature engineering • Pandas • SQL aggregation and filtering • Grouping and ranking • PostgreSQL • Dashboard interpretation • Business communication

Appendix: Business question index

No.	Business question
1	Which gender group contributes more purchase value?
2	How many discounted records have above-average purchase amounts?
3	Which five products have the highest average review ratings?
4	How do Standard and Express shipping compare by average purchase amount?
5	Do subscribed customers spend more?
6	Which products have the highest percentage of discount usage?
7	How many customers are New, Returning, and Loyal?
8	What are the three most purchased products within each category?
9	Among repeat buyers (more than five previous purchases), how many are subscribed?
10	What is the purchase-value contribution of each age group?